OPEN SOURCE NIGHT

Nightshift

From Copilot to Autopilot

Engineering guardrails for AI coding agents

Drew Schillinger
Enterprise Architect and community organizer

20 October 2026

Open Source Night at Tech Square ATL

OUR PATH

- Overview
- Skills
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- Routing
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Joint Atlanta developer-community meetup

OPEN SOURCE

An inspectable engineering workflow

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A useful starting point

Read the specification and tests.

Understand what the change should do.

An inspectable contribution

Show the source diff and evidence.

Let another reviewer challenge the result.

> The repository exposes the instructions as well as the implementation.

AGENTS.md; commands/; agents/; scripts/

OVERVIEW

The harness keeps the workflow

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Entry point

A skill or CLI starts the work.

Canonical instructions

Stages and roles define the work.

Routing

Configuration selects the provider.

Controller

Evidence determines the next stage.

AGENTS.md; skills/nightshift/SKILL.md; routing.json; scripts/nightshift-pipeline.py

The skill is the entry point

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CODEX SKILL

```
$nightshift <ticket-ref>
```

The adapter loads the canonical stage instructions.

A ticket starts the engineering pipeline without separate calls for each stage.

skills/nightshift/SKILL.md

SKILLS

The instructions stay in the repository

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commands/	Canonical stage instructions
agents/	Reusable role instructions
scripts/	Execution and validation
routing.json	Provider and model selection

AGENTS.md; commands/nightshift-product.md; agents/nightshift-spec-writer.md

Six stages retain their evidence

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Product

Specification and scope

2

Adversarial

Independent challenge

3

Implementation

Tests and source changes

4

Review

Independent code review

5

Drift

Specification alignment

6

QA

Final verification

commands/nightshift-{product,adversarial,implement,review,drift,qa}.md

A small task makes proof visible

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PUBLIC DURATION FORMATTER

```
format_seconds(0)    == "0:00"
format_seconds(125)  == "2:05"
format_seconds(3600) == "60:00"
```

Acceptance cases

Nonnegative integers format correctly.

Negative integers raise an error.

Booleans and other types raise an error.

Public rehearsal: duration.py; tests/test_duration.py; docs/133/IMPLEMENTATION.md

Routing selects the worker

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Author path

Codex

`gpt-6.1-sol`

Product and Implementation

Independent review path

Claude

`sonnet`

Adversarial, Review, Drift and QA

> These are the recorded routes for this acceptance run.

routing.json; accepted controller receipts for issue 133

Spend for the work being done

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Different tasks

Source lookup, implementation and adversarial review ask for different capabilities.

Total work cost

Compare context, tool support, retries, latency and acceptance—not just one response.

> The cheapest useful route is the one that produces accepted work with less total effort.

routing.json; scripts/nightshift-handoff.py; scripts/nightshift-ticket-budget.py

Challenge the author’s assumptions

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Author

Builds from the spec and can become attached to its own interpretation.

Recorded author: Codex.

Independent reviewer

Checks the source, tests and contract instead of accepting the author's summary.

Recorded reviewer: Claude.

> A second provider adds a different perspective. Shared blind spots can still survive.

routing.json; scripts/nightshift-agent.sh; docs/133/IMPLEMENTATION.md

Trust the checks around each worker

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Before dispatch

Explicit policy, available runtime, scope and retained allowance.

At the boundary

Role contract, structured output and locally checked evidence.

Before delivery

Review, drift and QA; explicit human decisions where required.

> Switching models changes the worker. It does not relax the gate.

scripts/nightshift-provider-policy.py; contracts/; scripts/nightshift-pipeline.py; AGENTS.md

Admission happens before dispatch

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Provider capability

Required runtime and tools are available.

Local inventory

The configured local model exists before extraction starts.

Retained allowance

Existing call and active-time limits still apply.

Bounded context

Stage handoffs carry relevant files and findings.

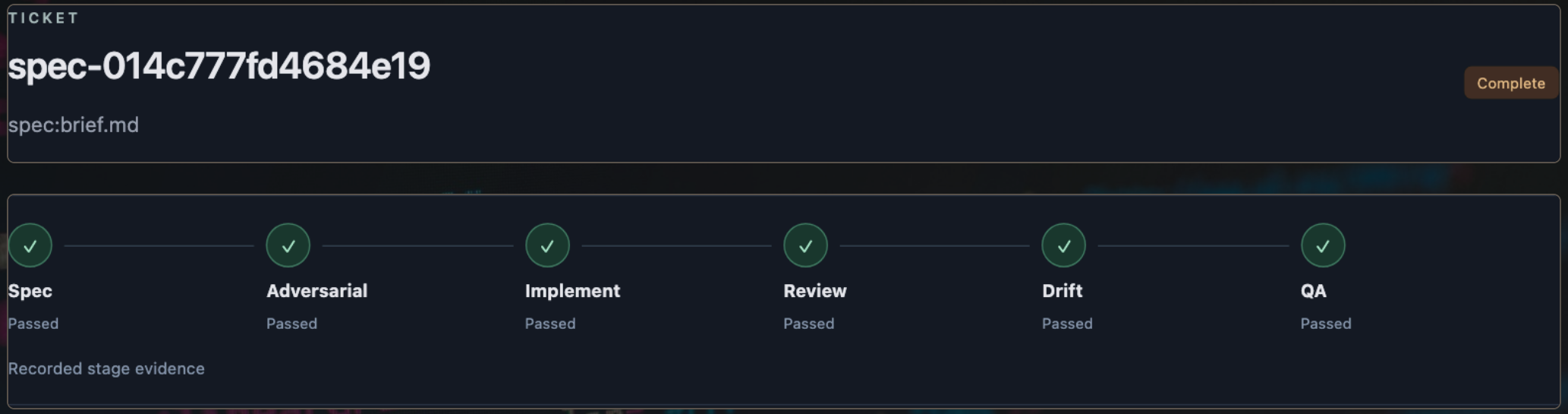
scripts/nightshift-capability.sh; scripts/nightshift-ticket-budget.py; scripts/nightshift-handoff.py

DEMONSTRATION

The front end shows the evidence

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Routes, receipts, artifacts and retained attempts remain inspectable.

Actual dashboard capture of the verified public rehearsal

Watch Nightshift work

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PRESENTER TERMINAL · FROM THE COMMUNITY CHECKOUT

```
python3 scripts/nightshift-talk-check.py
```

Recheck six stage receipts, real tests and both proof gates.

Repeat the retained task with zero new generation calls.

Open the local dashboard ↗ · Demo setup and commands ↗

Presenter laptop: retained fixture and subscription login required. Cached repeat, not a fresh agent run.

RED and GREEN remain inspectable

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Before implementation

Retained RED evidence records expected test failures.

Frozen cases define the required behavior.

After implementation

The real tests pass.

Development and final proof gates both pass.

> The rejected receipt stays in history.

commands/nightshift-implement.md; docs/133/IMPLEMENTATION.md

Zero generation calls on repeat

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6
accepted stages

0
additional generation calls

26
unchanged call reservations

Receipt hashes and attempt history stayed unchanged.

Current bindings allow the controller to reuse accepted work.

docs/133/VALIDATION.json; retained repeat-validation.json

EVIDENCE

Verification has a defined scope

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Observed

One public task passed all six stages.

Both proof gates and repeat reuse passed.

Ten CI shards and dashboard checks passed.

Still outside this evidence

Reliability across arbitrary projects

Actual provider billed charges

Hermes or Jev accuracy and savings

docs/133/VALIDATION.json; GitHub Actions run 38600812302

NIGHTSHIFT BUILDS NIGHTSHIFT

We pointed it at itself

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1

Build

A control-room UI for Nightshift.

2

Challenge

Review catches a hidden continuation control.

3

Repair

Restore access. Tests and re-review pass.

> It caught a regression—and exposed its own workflow bug.

Merged after operator approval.
The original run stopped at its work allowance.

#138 · Real run and evidence ↗ #139 · Human acceptance sent to code repair ↗ #140 · Operator-directed merge ↗

DEEP DIVE

Human checkpoints

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Declared operating mode

Supervised planning waits for approval.

Autonomous planning records its plan and continues.

Explicit boundaries

Production deployment requires confirmation.

Permanent removal requires confirmation.

> Passing a test does not grant new authority.

AGENTS.md; commands/nightshift-implement.md

FROM MY FIRST TALK

From memory to accountable work

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In Pursuit of Memory

Carry useful context between sessions.

Keep sources, identity and trust visible.

Nightshift

Turn context into a verifiable change.

Reduce repeated discovery and unnecessary work.

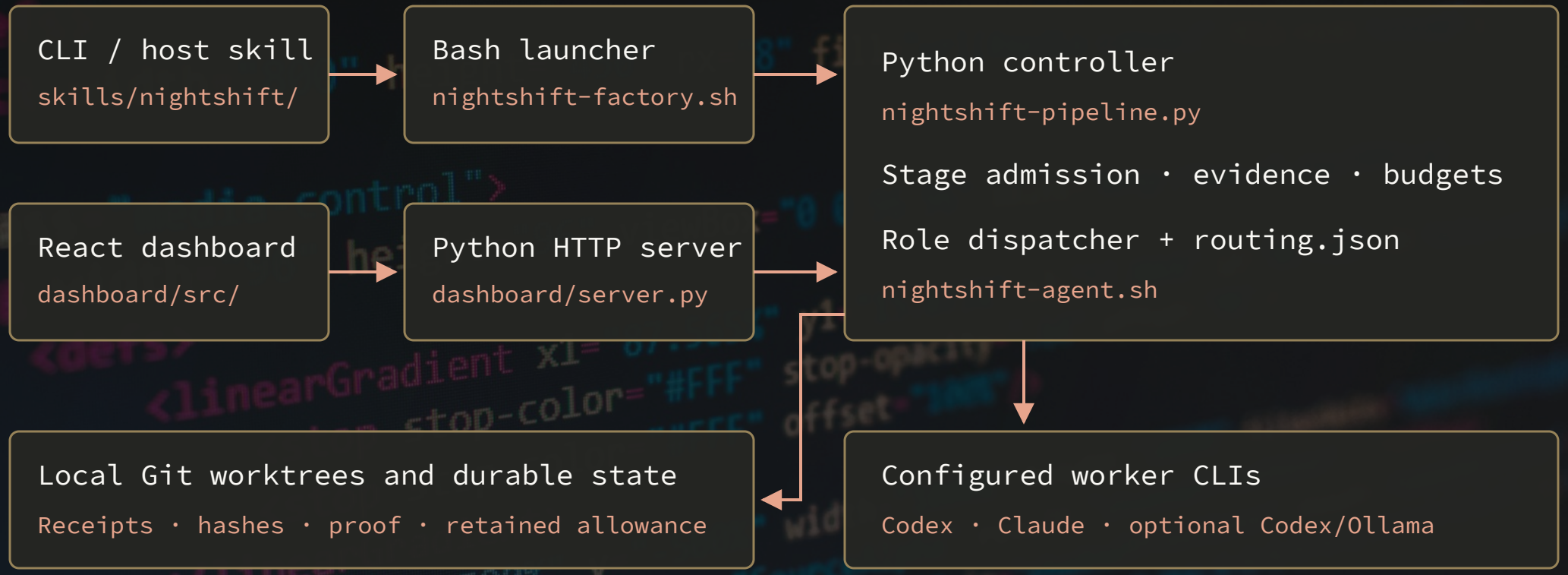
> Optimization goal: less rework per accepted change.

My first presentation ↗ · Optimization goals, not a measured savings claim

What runs where

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Control runs locally. Provider CLIs can reach remote models; the local-model route is optional.

A ticket becomes a verified change

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Product → Adversarial → Implement → Review → Drift → QA

Every stage → checked evidence → advance, repair or wait

Spec and scenarios · source and tests · receipts and hashes · persistent allowance

MEX · optional today

Bounded source context enters the stage handoff.

Missing or stale evidence stays unavailable.

Jev · shadow adapter

Advisory judgments beside required review.

Off in this demo; wider routing use is proposed.

Human checkpoints can pause a stage. Delivery: verified branch / optional PR; production approval is separate.

commands/ · agents/ · contracts/ · scripts/nightshift-{pipeline,handoff,jev-evaluation}.py

Context is evidence, not authority

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Untrusted input

Tickets and retrieved text cannot grant tool permissions.

Least privilege

Enforce scope at the tool and runtime boundary.

Controlled egress

Review the payload before sending private evidence.

Independent gates

Keep review, proof and human decisions distinct.

> A confidence score cannot authorize a consequential action.

AGENTS.md; scripts/nightshift-agent.sh; dashboard/server.py

Security lives below the prompt

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Transport

Verify the peer and its certificate.

Identity

Identify the caller; enforce authorization.

Data scope

Send only the evidence the task needs.

Retrieval is one part of memory

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Vector retrieval

Find candidate context by similarity.

A relevant match still needs a source.

Memory management

Track provenance, revision and freshness.

Decide what to retain, share and retire.

> Remember the reason—and the evidence it depends on.

MEX: carry the team’s context

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Repository memory

Architecture, decisions and handoffs through Git.

Code-linked grounding and a local graph.

My contribution

I contributed the C# code-graph extractor.

A concrete open-source entry point for .NET engineers.

Nightshift can retrieve bounded source context when the graph is available.

MEX ↗ · My C# contribution ↗ · scripts/nightshift-handoff.py

Ask a smaller decision

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Candidate uses

Classify a narrow claim.

Suggest a route or relevant context.

Return typed choices and confidence.

Nightshift boundary

Optional shadow adapter exists.

Disabled in this demonstration.

Required independent review stays.

> First measure decisions. Then decide what to delegate.

Jev community decision workflows ↗ · scripts/nightshift-jev-evaluation.py

COST AND OPTIMIZATION

Measure cost per accepted change

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Context

Retrieve and hand off the relevant evidence.

Routing

Match the worker to the actual task.

Reuse

Reuse accepted work while bindings match.

Quality

Count rework and missed defects too.

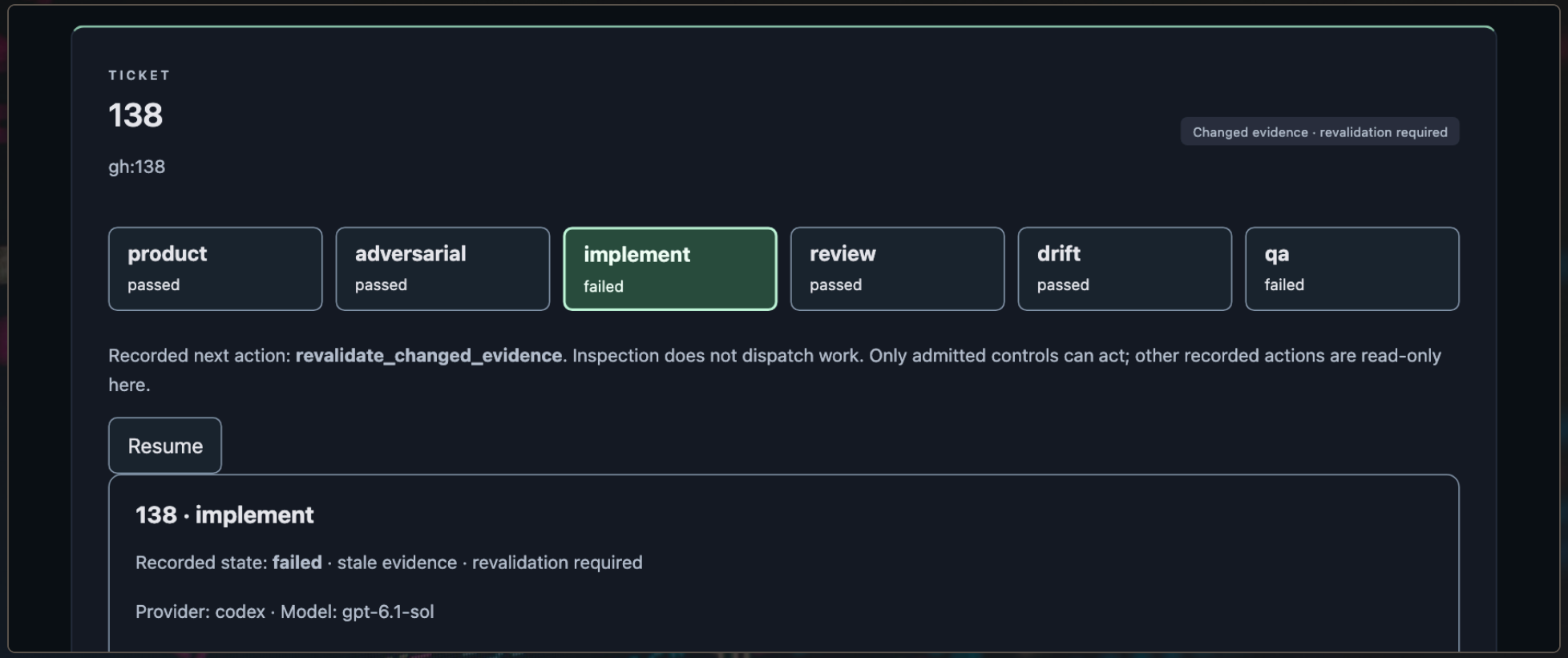
Track tokens, calls, elapsed time, retries and acceptance.
Reported estimates and actual billed cost are different measures.

scripts/nightshift-handoff.py; scripts/nightshift-ticket-budget.py; docs/talks/TESTING.md

The current control room

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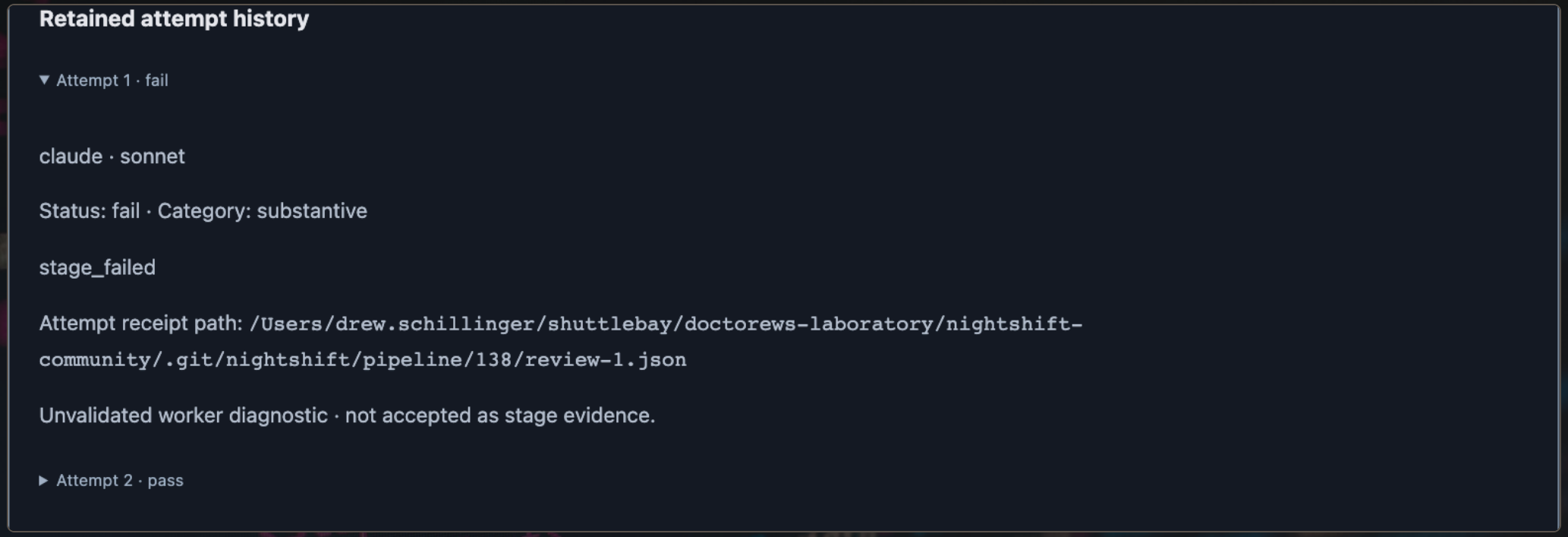
Select a stage → inspect its worker, evidence and history. This retained run needs revalidation.

STAGE EVIDENCE · ACTUAL LOCAL CAPTURE

Review keeps the failed attempt

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The first review found a hidden control. The repair passed the next review; earlier evidence remains.

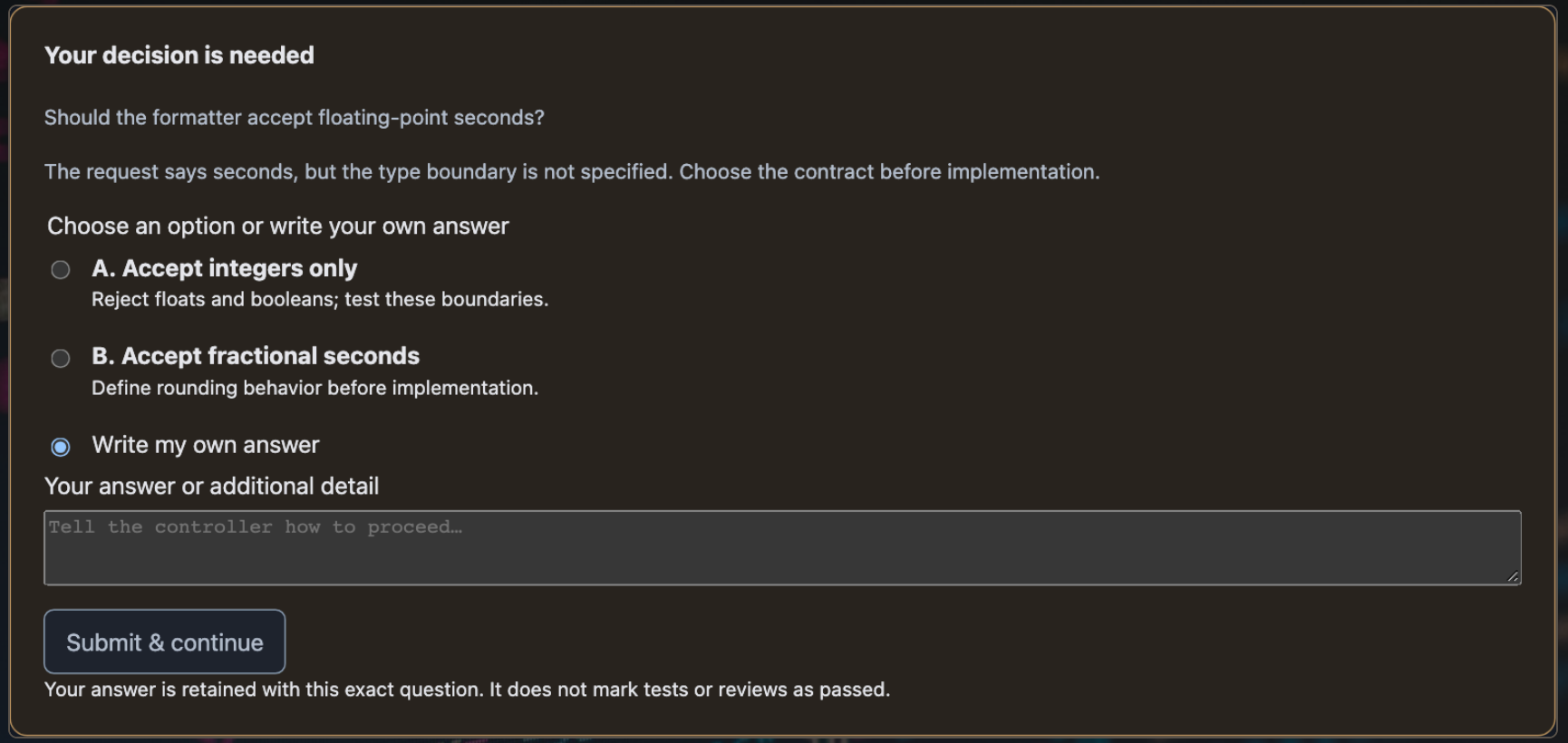
Actual task 138 Review panel; docs/138/IMPLEMENTATION.md; docs/138/REVIEW.md

HUMAN IN THE LOOP · ILLUSTRATIVE QUESTION

Answer the missing requirement

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Real UI with illustrative data. No answer submitted and no run started.

dashboard/src/ticket-live.jsx · TicketDecision; scripts/nightshift-console-decisions.py

Approve a specific spec version

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Read the contract before continuing. Changed inputs require fresh approval.

Illustrative data in actual UI; dashboard/src/app.jsx; scripts/nightshift-console-actions.py

A stopped worker is not a verdict

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Active time sums tracked invocations, including overlap. It is neither wall time nor a token bill.

Actual task 138 allowance; scripts/nightshift-ticket-budget.py; docs/138/OPERATOR-DELIVERY.md

Inside the Nightshift skill

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skills/nightshift/SKILL.md · EXCERPT

```
## Invocation
`$nightshift <ticket-ref>`
Advanced form: `$nightshift <stage> <arguments>`
Advisory commands: `help`, `explain`, `architect`,
`dev`, `pm`, `ux-designer`, `architecture`, `ux`, `bmad`.
```

One adapter loads shared instructions. Roles and provider selection live separately.

skills/nightshift/SKILL.md; commands/; agents/; routing.json

Follow one Product-stage handoff

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REPOSITORY FILES IN THE DISPATCH PATH

```
skills/nightshift/SKILL.md
  → commands/nightshift-product.md
  → agents/nightshift-code-fact-extractor.md
  → agents/nightshift-spec-writer.md
scripts/nightshift-agent.sh + routing.json
  → provider invocation + validated role output
```

The extractor grounds identifiers. The writer cites sources and records unknowns.

HARNESS · ACTUAL TEST EVIDENCE

Follow the formatter’s proof trail

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RETAINED PUBLIC TASK ARTIFACTS

```
SPEC.md + behavior-scenarios.json
  → implement-2-red.log   (expected failure)
  → proof-red.json        (development proof)
  → duration.py + tests/test_duration.py
  → proof-final.json      (final proof)
```

Three tests cover 26 inputs. Later review and QA inspect the resulting evidence.

Public formatter worktree docs/spec-014c777fd4684e19/; docs/133/IMPLEMENTATION.md

The answer does not pass the gate

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Exact question → Saved answer → Admission check → Resume or block

Retain the decision

Task, question and settings remain bound. Repeated questions can reuse the answer.

Retain the uncertainty

Changed evidence needs revalidation. Tests and reviews still decide their own outcomes.

scripts/nightshift-console-decisions.py; scripts/nightshift-console-actions.py

MEX retrieval inside a handoff

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COMMAND BUILT BY scripts/nightshift-handoff.py

```
mex graph scope "<task query>" --detail source \
  --max-files 3 --max-nodes 12 \
  --max-output-tokens 2000 --max-source-lines 60
```

Keep source-backed matches. Missing, stale or partial graph evidence becomes unavailable.

> Retrieved context can inform a stage; it cannot approve one.

What we would ask Jev

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EXISTING RUBRIC · scripts/nightshift-efficiency.py

```
supported:
  Does supplied evidence support its own claims?
limitations:
  Does it distinguish observations from unknowns?
Both questions: type = "noul"
```

Evaluate beside the reviewer first. Measure misses and false alarms before changing dispatch.

Prove an optimization before promotion

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1 · Baseline

Same public task, fixed acceptance cases and retained failures.

2 · Add context

Compare bounded MEX retrieval with existing handoffs.

3 · Shadow decisions

Compare Jev judgments with labelled cases and reviewer findings.

Promote only with acceptable defect detection, total work cost and latency. Preserve human boundaries.

Proposed evaluation plan; scripts/nightshift-jev-evaluation.py; docs/133/VALIDATION.json

A contribution people can review

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A clear behavior, a focused change and evidence someone else can inspect.

github.com/doctor-ew/nightshift-community

Questions and discussion

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Testing the retained demo

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FROM THE COMMUNITY CHECKOUT

```
python3 scripts/nightshift-talk-check.py
```

The checker validates identity, receipts, proof and real tests.

The repeat check refuses new provider generation.

scripts/nightshift-talk-check.py; docs/talks/TESTING.md

DEEP DIVE

SOLID and ACID in review

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SOLID

Responsibilities and interface contracts

Dependencies and extension points

ACID

Validation and consistent state

Isolation and recovery for state changes

> These are review lenses. Findings still need evidence.

commands/nightshift-review.md